NEURALPLANE: Structured 3D Reconstruction in Planar Primitives with Neural Fields

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01 Why Reconstruct Scenes With Planar Primitives?

Motivation And Problem Scope

Indoor scenes are dominated by meaningful planar surfaces

- Planes provide compact, expressive structure for walls, floors, furniture
- Facilitates crisp, structured 3D reconstructions

Traditional pipelines fit planes to noisy geometry

- Depend on high-quality point clouds/meshes
- Often decouple geometry from semantics

Limitations Of Existing Approaches

Geometry+RANSAC requires high-quality geometry

- Misses semantic boundaries and coplanar distinctions

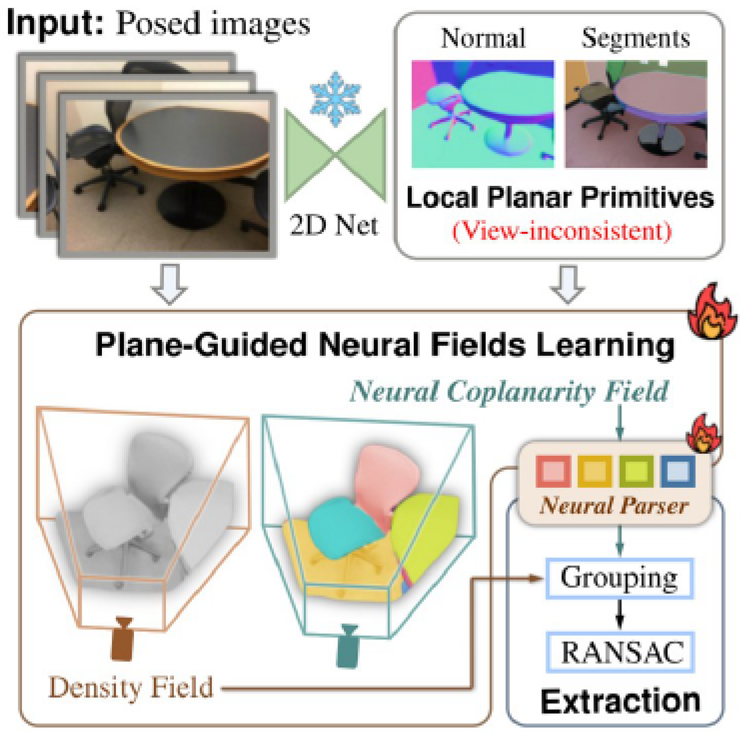
Learning-based plane mappers need labels or fail in complexity

- Struggle with complex indoor layouts without plane annotations

Single-view predictions are cross-view inconsistent

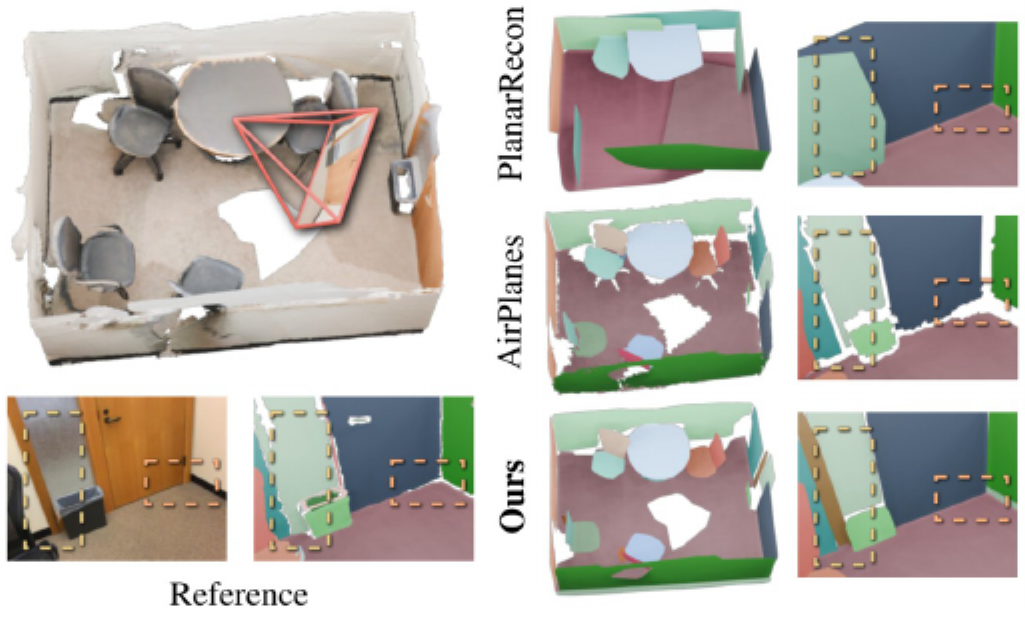
- Hinders reliable multi-view plane fusion

02 Core Ideas And Contributions



- Distill inconsistent 2D planes into unified 3D neural representation
  - Couples geometry and semantics without plane annotations
- Training-free monocular module for smooth plane segments
  - Outputs coarse parameters guiding later refinement
- Plane-guided density optimization with normal and pseudo-depth
  - Jointly refines plane parameters during training
- Neural Coplanarity Field with

02 Pipeline At A Glance



- Three-phase pipeline integrates 2D cues with neural fields
- Phase 1: local planar primitives per image
  - Monocular priors yield over-segmented, robust plane observations

03 Related Work And Positioning

Multi-View Plane Reconstruction

Prior methods detect planes from RGB or geometry

- Cross-view inconsistencies limit multi-view fusion

Incremental fusion and RANSAC improve precision

- Still hinge on geometry quality and 2D labels

NeuralPlane implicitly fuses plane cues in neural fields

- Leverages semantics without plane annotations

Neural Rendering For Higher-Level Mapping

Neural fields encode sparse primitives and semantics

- Via feature distillation and contrastive learning

NeuralPlane jointly models planar geometry and semantics

- Density encodes surfaces; NCF captures coplanarity features

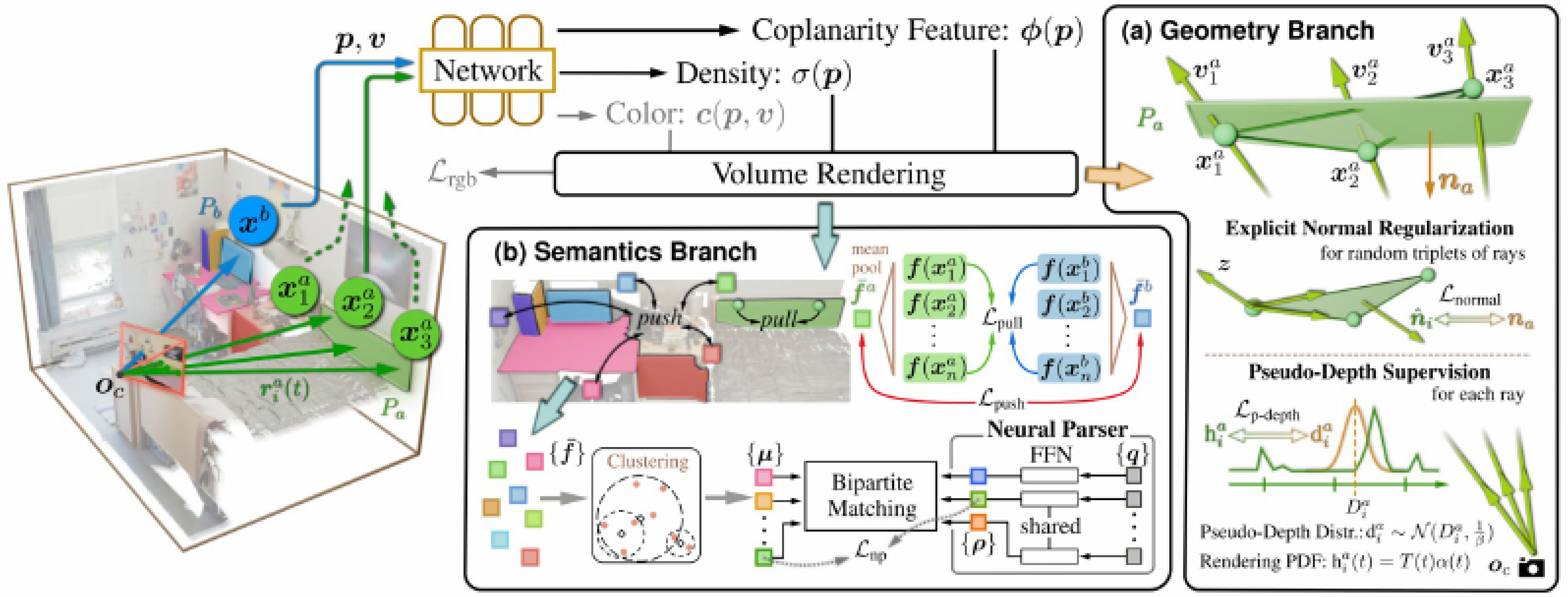
Enables instance-consistent plane extraction

- Bridges low-level rendering and high-level mapping

04 Problem Setup And Representation

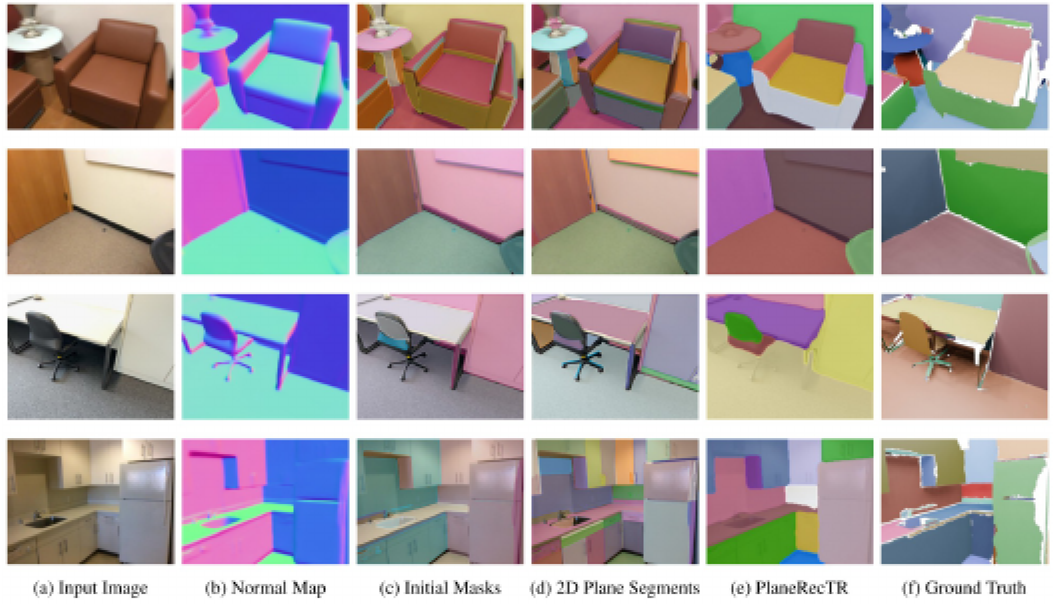
- Input: posed images; output: 3D planar primitives
- Local primitives P=(M⊆I, π=[n,o]) from single images
  - n unit normal; o plane offset
- Neural radiance field for density/color; auxiliary NCF
  - Coplanarity-aware features rendered along rays
- Training: photometric plus plane-guided geometry/semantics losses

04 Three-Phase Workflow



- Phase 1: derive local plane segments using monocular priors
- Phase 2: train neural fields with plane guidance
  - Geometry: normal, pseudo-depth, parameter refinement

05 Plane Segments From Monocular Priors



- Predict normals, cluster, and refine with SAM masks
- Select large masks as 2D plane segments
  - Over-segmented yet robust observations

05 Initializing Local Plane Geometry

- Plane normal: mean predicted normal within segment
- Plane offset: weighted fit to SfM keypoints
  - Weights inverse to average reprojection error
- Coarse parameters guide training-time refinement

06 Phase II: Plane-Guided Neural Fields Learning

Geometry Branch: Explicit Normal Regularization

Sample ray triplets within plane segments

- Compute algebraic plane normals from termination points

Penalize deviation from initialized normals

- Enforces cross-region normal consistency

Promotes smooth, accurate planar surfaces

Geometry Branch: Pseudo-Depth Supervision And Refinement

Compute pseudo-depth from plane parameters and rays

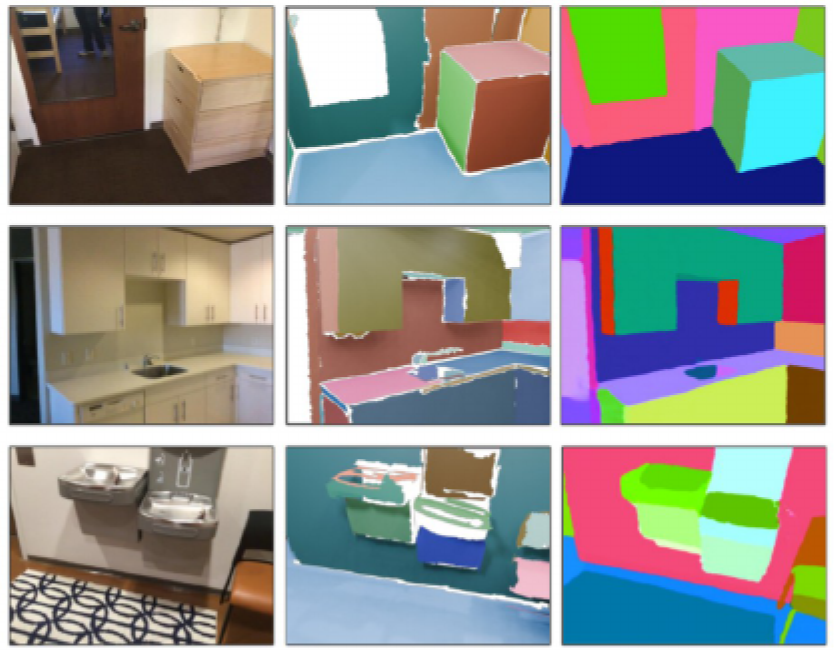
Model uncertainty with Gaussian; minimize KL to PDF

- Robust alignment to rendered distributions

Backpropagate to refine normals and offsets

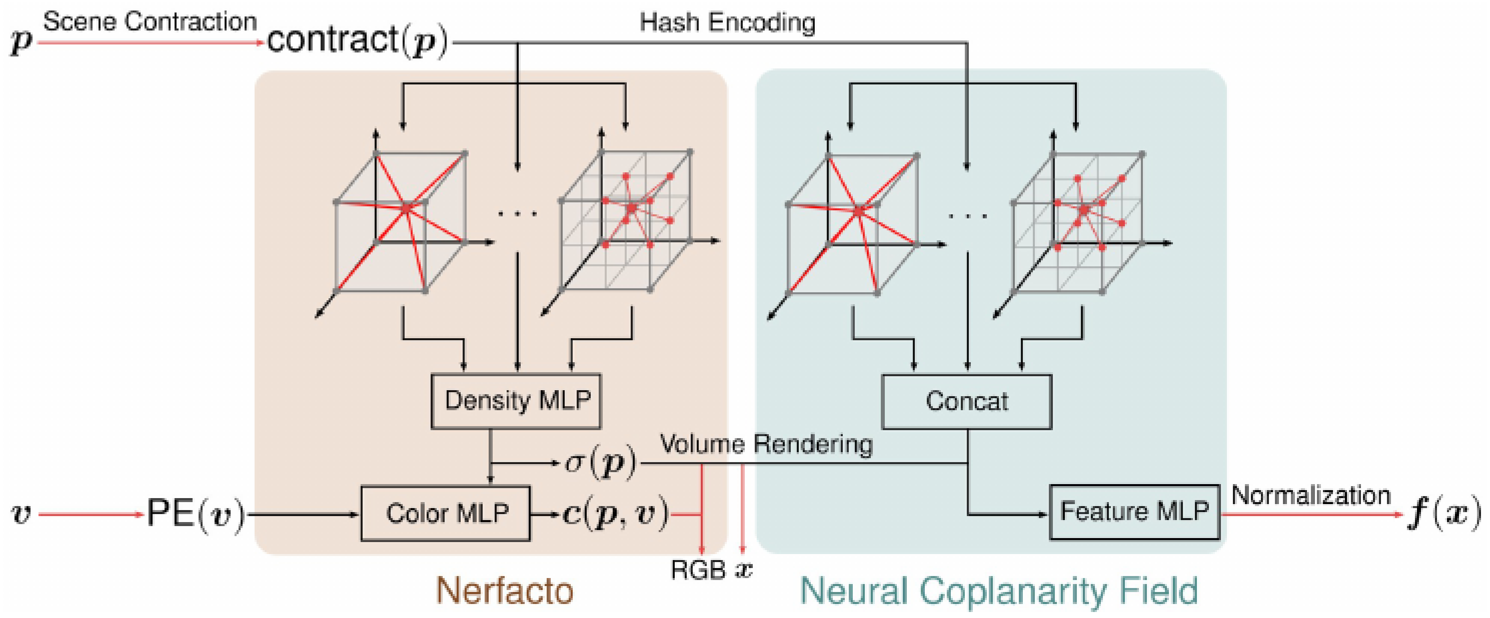
- Improves geometry beyond normal smoothness

06 Semantics Branch: Neural Coplanarity Field (NCF)



- Define d-dimensional feature field rendered along rays
- Contrastive learning pulls same-plane features together
- Push apart primitives differing in normals/offsets
  - Thresholded by coplanarity criteria
- Captures coplanarity-aligned semantic boundaries

06 Semantics Branch: Neural Parser For Prototypes

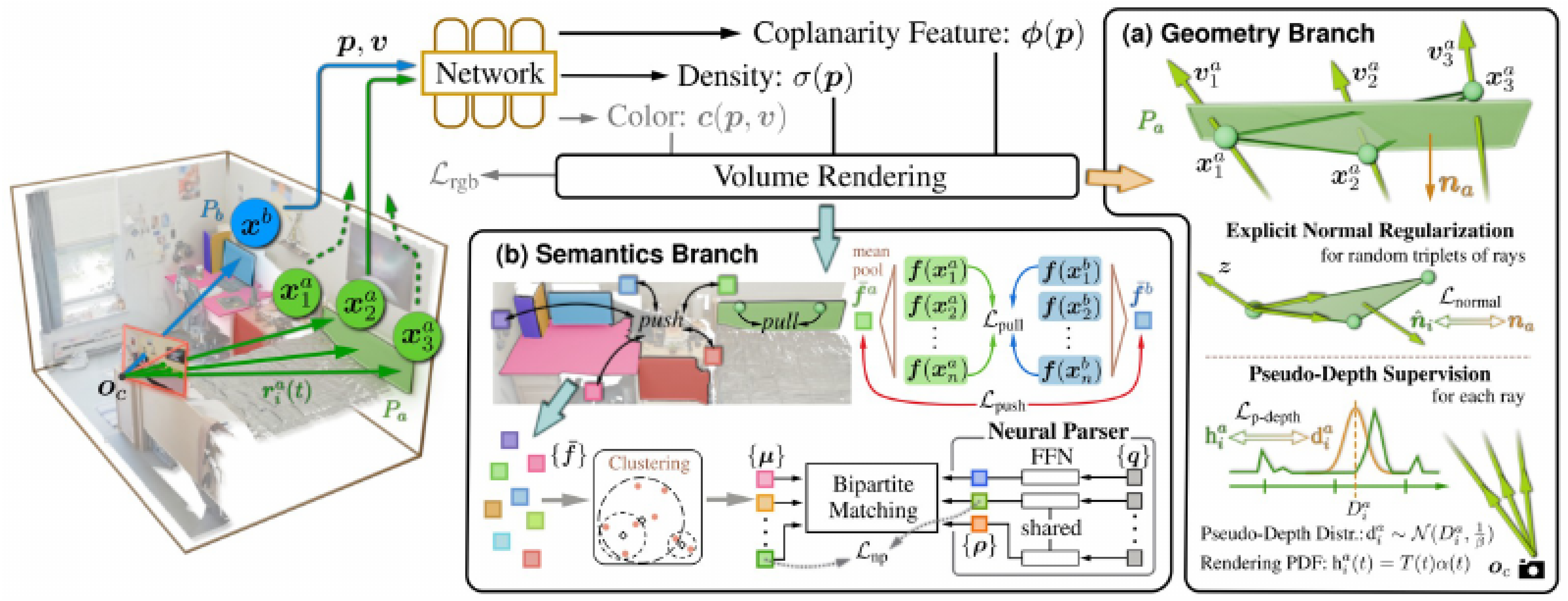


- Query-based prototypes predicted from learnable queries
- Cluster features with DBSCAN; compute centroids
- Hungarian matching assigns centroids to prototypes

06 Overall Objective And Training Schedule

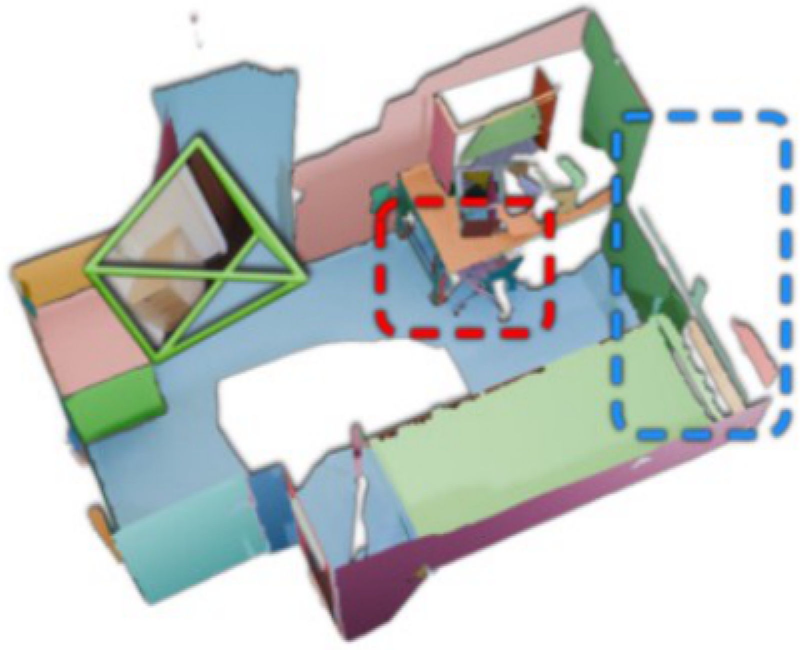
- Objective: weighted sum of photometric and plane-guided losses
  - Normal, pseudo-depth, NCF pull/push, parser terms
- Schedule: early suppress unreliable primitives
  - Few keypoints deferred; later re-estimate offsets
- Activate full losses for joint refinement
  - Fixed hyperparameters across scenes

07 Two-Stage Plane Extraction Procedure



- Render dense termination points and NCF features
- Assign to nearest learned prototype; group by labels
- Fit parametric planes via RANSAC

08 Datasets And Baselines



- Evaluate on 8 ScanNetv2 and 4 ScanNet++ indoor scenes
- Compare against geometry+RANSAC and learning baselines
  - COLMAP, SimpleRecon, FineRecon, ManhattanSDF, NeuRIS, MonoSDF, ObjectSDF++
- Include AirPlanes and Sequential RANSAC variants
- Variant with PlaneRecTR for local primitives

08 Metrics And Implementation

Metric	Definition
Accuracy	$\text{mean}_{p \in \mathbb{P}} \left(\min_{p^* \in \mathbb{P}^*} \lVert p - p^* \rVert_2 \right)$
Completeness	$\text{mean}_{p^* \in \mathbb{P}^*} \left(\min_{p \in \mathbb{P}} \lVert p - p^* \rVert_2 \right)$
Chamfer	$(\text{Accuracy} + \text{Completeness})/2$
Precision (%)	$\text{mean}_{p \in \mathbb{P}} \left(\min_{p^* \in \mathbb{P}^*} \lVert p - p^* \rVert_2 < 0.05 \right) \times 100$
Recall (%)	$\text{mean}_{p^* \in \mathbb{P}^*} \left(\min_{p \in \mathbb{P}} \lVert p - p^* \rVert_2 < 0.05 \right) \times 100$
F-score (%)	$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$

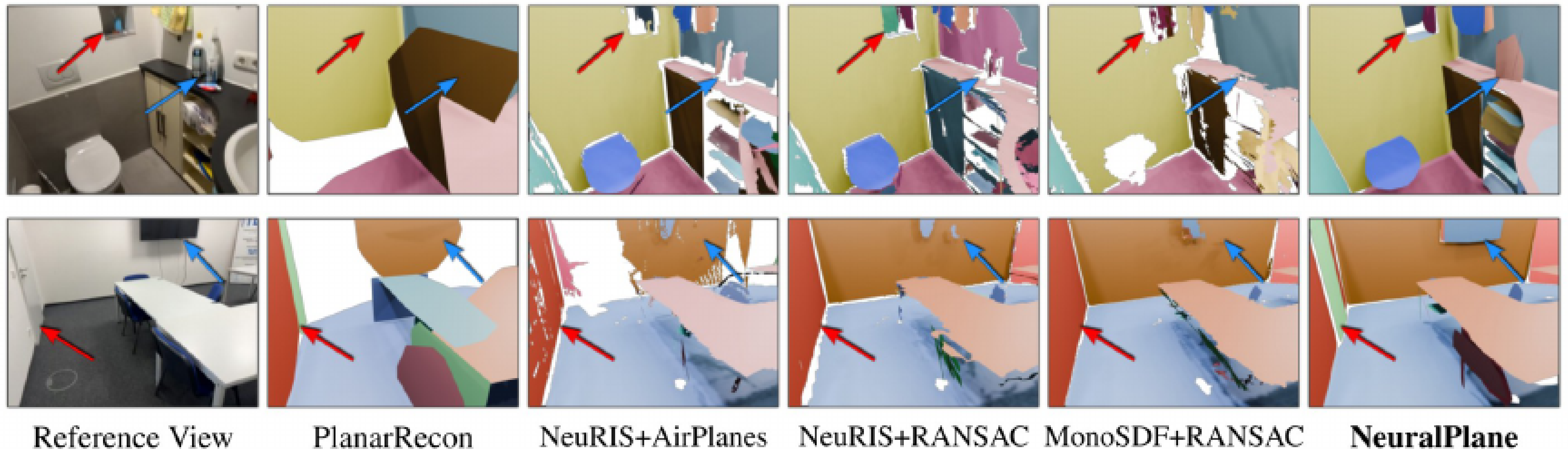
- Geometry: Accuracy, Completeness, Chamfer, Precision, Recall, F-score
- Segmentation: RI, VOI, SC
- Implementation: Nerfstudio (Nerfacto), ~4k iters, batched rays

09 Quantitative And Qualitative Findings On ScanNetv2

Method		Geometry: Accu. ↓	Comp. ↓	**Chamfer** ↓	Prec. ↑	Recall ↑	**F-score** ↑	Segmentation: RI ↑	VOI ↓	SC ↑
COLMAP (Schönberger et al., 2016)		18.35	13.40	**15.88**	40.5	42.4	**41.2**	-	-	-
SimpleRecon (Sayed et al., 2022)		6.74	5.55	**6.15**	59.2	58.8	**59.0**	-	-	-
FineRecon (Stier et al., 2023)		6.12	4.19	**5.16**	68.6	72.9	**70.6**	-	-	-
ManhattanSDF (Guo et al., 2022)		9.95	7.51	**8.73**	50.4	50.9	**50.6**	-	-	-
NeuRIS (Wang et al., 2022)		10.23	5.69	**7.96**	62.0	64.8	**63.2**	-	-	-
MonoSDF‡ (Yu et al., 2022b)		5.28	5.07	**5.18**	69.7	69.7	**69.7**	-	-	-
ObjectSDF++† (Wu et al., 2023a)		8.82	5.25	**7.03**	54.9	68.9	**60.9**	-	-	-
COLMAP	+Seq.RANSAC	19.32	13.37	**16.34**	40.6	40.9	**40.6**	0.928	3.91	0.154
SimpleRecon	+Seq.RANSAC	6.24	6.00	**6.12**	57.7	52.6	**54.9**	0.949	2.65	0.272
FineRecon	+Seq.RANSAC	5.20	5.65	**5.43**	69.1	64.9	**66.7**	0.941	2.56	0.276
ManhattanSDF	+Seq.RANSAC	9.37	8.70	**9.04**	50.6	51.0	**50.8**	0.930	2.82	0.281
NeuRIS	+Seq.RANSAC	9.87	6.35	**8.11**	59.6	59.3	**59.3**	0.945	2.57	0.293
MonoSDF	+Seq.RANSAC	5.91	5.43	**5.67**	65.9	66.1	**65.9**	0.945	2.38	0.333
ObjectSDF++‡	+Seq.RANSAC	8.41	5.35	**6.88**	58.0	68.8	**62.8**	0.952	2.32	0.334
SimpleRecon	+AirPlanes (Watson et al., 2024)	5.43	6.60	**6.01**	59.2	51.5	**55.1**	0.944	2.51	0.341
FineRecon	+AirPlanes (Watson et al., 2024)	4.93	5.95	**5.44**	70.8	62.2	**66.2**	0.947	2.43	0.310
ManhattanSDF	+AirPlanes (Watson et al., 2024)	9.46	9.30	**9.38**	52.7	50.2	**51.3**	0.940	2.61	0.315
NeuRIS	+AirPlanes (Watson et al., 2024)	6.05	7.38	**6.71**	66.2	56.7	**61.0**	0.943	2.55	0.291
MonoSDF	+AirPlanes (Watson et al., 2024)	4.57	6.33	**5.45**	72.2	62.0	**66.6**	0.948	2.38	0.346
PlanarRecon (Xie et al., 2022)		7.74	11.85	**9.80**	55.2	44.3	**49.0**	0.909	3.27	0.265
NeuralPlane@PlaneRecTR		5.38	4.65	**5.02**	67.6	70.0	**68.7**	0.949	2.37	0.364
NeuralPlane		4.92	4.27	**4.59**	70.5	71.9	**71.2**	0.955	2.25	0.376

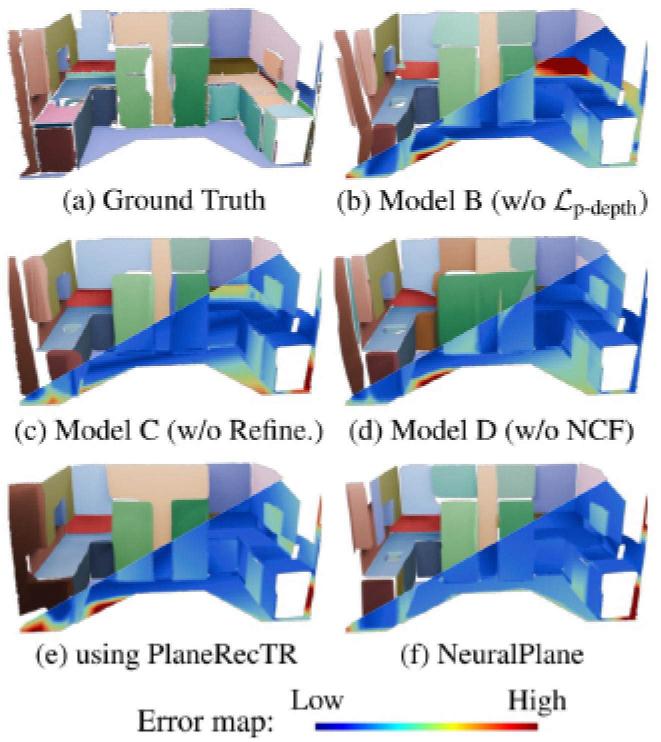
- Achieves top Chamfer and F-score; best segmentation metrics
- AirPlanes variants: high precision, lower recall
  - Indicates missed planes

09 Generalization To Low-Texture ScanNet++



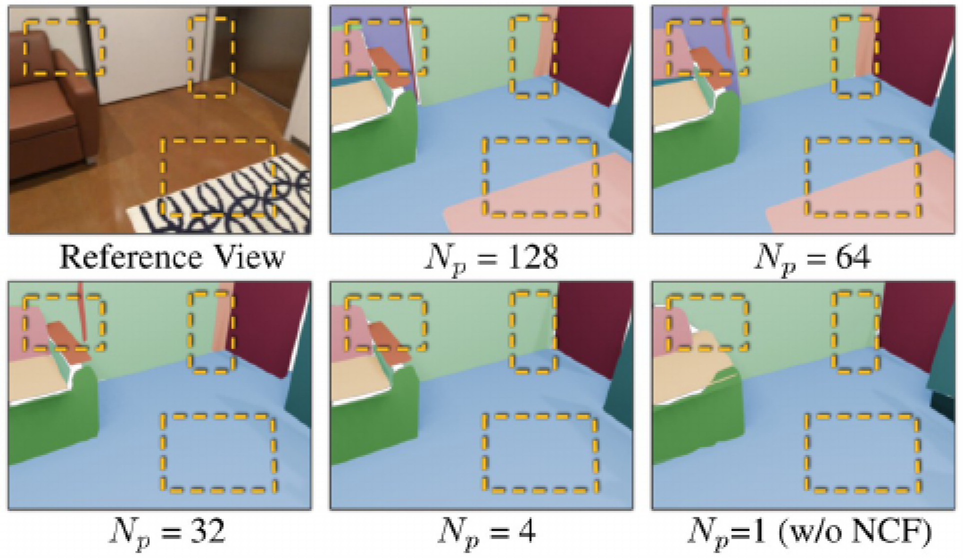
- Competitive geometry and strong segmentation on low-texture data
- Preserves fine-grained semantics despite sparse keypoints
- Balances priors and efficiency via density representation

10 Component-wise Ablation



- Normal or pseudo-depth removal degrades geometry
- Skipping parameter refinement hurts accuracy
- Omitting semantics induces conflicts between coplanar regions
- Full model yields best geometry and segmentation
  - Demonstrates geometry–semantics synergy

10 Effect Of Prototype Count



(b) Different Levels of Detail

- Prototype count controls segmentation granularity
- Too many prototypes slightly reduce metrics
  - Over-segmentation risk

Summary And Outlook

- NeuralPlane unifies geometry and semantics without plane annotations
- Monocular priors guide density learning; NCF and Parser separate semantics
- Extraction yields crisp, coherent planar primitives
- Future: adaptive LOD, broader scenes, robotics/AR integration

THANKS!